

<https://github.com/Nicholas-Kang/5393HW2-git.git>

1. For the first problem, I started off with my own code and then had AI help me fix the code to output the correct sequence. I understood the problem more by having AI give me examples and to show me how it works. I had it improve my code and finalize a working version.

```
PS C:\Users\kangl\AppData\Local\Programs\Microsoft VS Code> & C:\Users\kangl\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/kangl/Downloads/5393HW2_P1.py
Testing starting values: 0, 1
Initial State: A=0, B=1
Sequence Step 2: A= 1, B= 1 (Current Value: 1)
Sequence Step 3: A= 1, B= 2 (Current Value: 2)
Sequence Step 4: A= 2, B= 3 (Current Value: 3)
Sequence Step 5: A= 3, B= 5 (Current Value: 5)
Sequence Step 6: A= 5, B= 8 (Current Value: 8)
Sequence Step 7: A= 8, B= 13 (Current Value: 13)
Sequence Step 8: A= 13, B= 21 (Current Value: 21)
Sequence Step 9: A= 21, B= 34 (Current Value: 34)
Sequence Step 10: A= 34, B= 55 (Current Value: 55)
Sequence Step 11: A= 55, B= 89 (Current Value: 89)
Sequence Step 12: A= 89, B=144 (Current Value: 144)
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Testing starting values: 3, 7
Initial State: A=3, B=7
Sequence Step 2: A= 7, B= 10 (Current Value: 10)
Sequence Step 3: A= 10, B= 17 (Current Value: 17)
Sequence Step 4: A= 17, B= 27 (Current Value: 27)
Sequence Step 5: A= 27, B= 44 (Current Value: 44)
Sequence Step 6: A= 44, B= 71 (Current Value: 71)
Sequence Step 7: A= 71, B=115 (Current Value: 115)
Sequence Step 8: A=115, B=186 (Current Value: 186)
Sequence Step 9: A=186, B=301 (Current Value: 301)
Sequence Step 10: A=301, B=487 (Current Value: 487)
Sequence Step 11: A=487, B=788 (Current Value: 788)
Sequence Step 12: A=788, B=1275 (Current Value: 1275)
-----
```

Testing starting values: 0, 1

Initial State: A=0, B=1

Sequence Step 2: A= 1, B= 1 (Current Value: 1)
Sequence Step 3: A= 1, B= 2 (Current Value: 2)
Sequence Step 4: A= 2, B= 3 (Current Value: 3)
Sequence Step 5: A= 3, B= 5 (Current Value: 5)
Sequence Step 6: A= 5, B= 8 (Current Value: 8)
Sequence Step 7: A= 8, B= 13 (Current Value: 13)
Sequence Step 8: A= 13, B= 21 (Current Value: 21)
Sequence Step 9: A= 21, B= 34 (Current Value: 34)
Sequence Step 10: A= 34, B= 55 (Current Value: 55)
Sequence Step 11: A= 55, B= 89 (Current Value: 89)
Sequence Step 12: A= 89, B=144 (Current Value: 144)

Testing starting values: 3, 7

Initial State: A=3, B=7

Sequence Step 2: A= 7, B= 10 (Current Value: 10)

Sequence Step 3: A= 10, B= 17 (Current Value: 17)
 Sequence Step 4: A= 17, B= 27 (Current Value: 27)
 Sequence Step 5: A= 27, B= 44 (Current Value: 44)
 Sequence Step 6: A= 44, B= 71 (Current Value: 71)
 Sequence Step 7: A= 71, B=115 (Current Value: 115)
 Sequence Step 8: A=115, B=186 (Current Value: 186)
 Sequence Step 9: A=186, B=301 (Current Value: 301)
 Sequence Step 10: A=301, B=487 (Current Value: 487)
 Sequence Step 11: A=487, B=788 (Current Value: 788)
 Sequence Step 12: A=788, B=1275 (Current Value: 1275)

2. I used AI to help me with 2 different versions of code. The first was a mathematical calculation to generate the outputs for each cycle. I had it help me write the model for this problem and to understand how the problem is supposed to be solved. The second method was to use the Aleae model as told from the problem. I had it help me write a reactions.txt for the filter topology and state.txt for the molecule counts and delay for each cycle, and then I tracked the results. I did this by using the command ./aleae state.txt reactions.txt 1 15 9 for each cycle.

Mathematic Sim Results

```
PS C:\Users\kangl\AppData\Local\Programs\Microsoft VS Code> & C:\Users\kangl\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/kangl/Downloads/5393HW2_P2_v2.py
Cycle 1: Input X = 100 | Output Y = 12.5000
Cycle 2: Input X = 5 | Output Y = 14.6875
Cycle 3: Input X = 500 | Output Y = 79.0234
Cycle 4: Input X = 20 | Output Y = 77.3389
Cycle 5: Input X = 250 | Output Y = 115.7953
```

```
PS C:\Users\kangl\AppData\Local\Programs\Microsoft VS Code> & C:\Users\kangl\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/kangl/Downloads/5393HW2_P2_v2.py
Cycle 1: Input X = 100 | Output Y = 12.5000
Cycle 2: Input X = 5 | Output Y = 14.6875
Cycle 3: Input X = 500 | Output Y = 79.0234
Cycle 4: Input X = 20 | Output Y = 77.3389
Cycle 5: Input X = 250 | Output Y = 115.7953
```

Aleae Sim results

Cycle 1: 14
 Cycle 2: 17
 Cycle 3: 65
 Cycle 4: 5
 Cycle 5: 36

Cycle 1 results:

trial stats:

events 12600

time inf

runtime 49.99.16

simulation stats:

avg [1.0000, 14.0000, 0.0000, 4.0000, 70.0000, 18.0000, 8.0000, 0.0000, 5.0000, 7.0000,
7.0000, 1.0000, 1.0000, 270.0000, 191.0000, 1021.0000, 11.0000, 49.0000, 0.0000, 1000.0000,
1000.0000, 1000.0000, 1036.0000]

avg events/trial 12600.0000

avg time/trial inf

avg runtime 49.101.388

total runtime 49.101.388

Output y = 14.000

Cycle 2 Results:

events 12600

time inf

runtime 43.742.397

simulation stats:

avg [2.0000, 17.0000, 12.0000, 49.0000, 1.0000, 3.0000, 3.0000, 1.0000, 7.0000, 7.0000,
7.0000, 10.0000, 10.0000, 662.0000, 1155.0000, 247.0000, 25.0000, 0.0000, 14.0000,
1000.0000, 1000.0000, 1000.0000, 1538.0000]

avg events/trial 12600.0000

avg time/trial inf

avg runtime 43.745.714

total runtime 43.745.714

Output Y = 17.0

Cycle 3:

events 12600

time inf

runtime 42.705.648

simulation stats:

avg [0.0000, 65.0000, 497.0000, 56.0000, 13.0000, 0.0000, 0.0000, 0.0000, 5.0000, 7.0000,
7.0000, 7.0000, 13.0000, 1355.0000, 92.0000, 358.0000, 0.0000, 23.0000, 1000.0000,
1000.0000, 1000.0000, 7448.0000]
avg events/trial 12600.0000
avg time/trial inf
avg runtime 42.708.756
total runtime 42.708.756

Output Y = 65.000

Cycle 4:

events 12600
time inf
runtime 41.1.47

simulation stats:

avg [1.0000, 5.0000, 382.0000, 47.0000, 137.0000, 21.0000, 0.0000, 0.0000, 2.0000, 3.0000,
3.0000, 7.0000, 7.0000, 21.0000, 1391.0000, 275.0000, 254.0000, 13.0000, 1.0000, 1000.0000,
1000.0000, 1000.0000, 14594.0000]
avg events/trial 12600.0000
avg time/trial inf
avg runtime 41.3.358
total runtime 41.3.358

Output Y = 5.000

Cycle 5:

simulation stats:

avg [0.0000, 36.0000, 0.0000, 0.0000, 736.0000, 97.0000, 2.0000, 0.0000, 9.0000, 7.0000,
7.0000, 6.0000, 6.0000, 48.0000, 6.0000, 396.0000, 55.0000, 540.0000, 0.0000, 1000.0000,
1000.0000, 1000.0000, 19848.0000]
avg events/trial 12600.0000
avg time/trial inf
avg runtime 41.732.115
total runtime 41.732.115

Output Y: 36